

EDUCATION

University of Technology Sydney

B.Eng (Hons) / Dip. Professional Engineering

Practice

Major: Mechatronics Engineering

Jan 2024 - Nov 2029 (expected)

WAM 81.12GPA 5.92

St Mary's Cathedral College

High School Diploma

1st — Engineering Studies, Sydney Catholic

Schools

ATAR 88.9 | 2022–2023

RELEVANT COURSEWORK

Sensors & Control for Mechatronic	D · 82
Systems	
Embedded Mechatronics Systems	HD
Programming 1 (Python & fundamentals)	HD · 92
Industrial Robotics (6-DOF simulation & control)	D
Intro to Mechatronics Engineering	HD
Engineering Project Appraisal	D · 76

TECHNICAL SKILLS

SENSING & SIGNAL PROCESSING

- Sensor IntegrationTime-of-Flight
- Analogue FilteringHeart Rate Sensing
- Control SystemsPID

PROGRAMMING

- PythonC/Embedded CSTM32CubeIDE
- MATLAB

HARDWARE & EMBEDDED

- STM32 MicrocontrollersAltium PCB Design
- SMD Soldering (0402)Fusion 360
- SolidWorks

TOOLS & PLATFORMS

- Rapid PrototypingMicrosoft Azure AIGit
- MS Office

CERTIFICATIONS

Microsoft Azure AI Fundamentals

Microsoft · Jul 2024 · ID C10243EC6D0E908E

C Programming for Embedded Applications

LinkedIn Learning · Jul 2024

First Aid & CPR / BLS

Current certification

General Construction Induction (White Card)

SafeWork NSW

AWARDS

UTS Dean's List

University of Technology Sydney · 2024

Rookie of the Year

UTS Motorsports Autonomous · 2024

1st Place — Engineering Studies

Sydney Catholic Schools · 2023

PROFILE

Penultimate-year Mechatronics Engineering student with hands-on experience in embedded sensing systems, analogue signal processing, and rapid hardware prototyping. Passionate about applying sensor fusion and AI to solve real-world infrastructure challenges. Proven ability to design, build, and iterate on multi-sensor hardware from PCB layout through to software integration — independently and as part of high-performing research teams.

RELEVANT EXPERIENCE

Product Development Engineer

"I did a thing" · YouTube5.6M+ SUBSCRIBERS

May 2025 - Present

- Sole engineer delivering 10+ functional prototypes for one of YouTube's largest engineering channels — averaging just 30 working hours per complete project cycle (ideation → sourcing → fabrication → iteration).
- Designed and built a wearable flotation device with integrated sail mechanism; produced an autonomous CO₂-actuated projectile system — both requiring rigorous sensor and actuator integration.
- Full engineering autonomy across every stage; all prototypes built to withstand hours of repeated takes during professional filming.

Embedded Systems Engineer

UTS Motorsports Autonomous

Apr 2024 - Aug 2025

- Developed real-time embedded firmware in C for STM32 microcontrollers on an autonomous Formula SAE vehicle, including sensor data acquisition and actuation control.
- Contributed to steer-by-wire system development; showcased technology at SXSW Sydney 2024 to industry professionals, receiving recognition as Rookie of the Year.
- Collaborated within a multidisciplinary team of 30+ across mechanical, electrical, and software subsystems — analogous to the UTS TRU / nbn multi-stakeholder structure.

Casual Academic — Mechatronics Engineering

University of Technology Sydney

Feb 2025 - Present

- Tutored undergraduate cohorts in 41099 Introduction to Mechatronics Engineering, reinforcing concepts in sensor systems, embedded programming, and control theory.

Avionics Trainee

UTS Rocketry Team

Apr 2024 - Jan 2025

- Prototyped avionics data loggers to capture and record in-flight sensor data (accelerometers, altimeters); attended workshops on flight data processing and analysis.

KEY TECHNICAL PROJECTS

Multi-Sensor PCB — Step Tracker & Sensing

HD — Embedded Mechatronics Studio 2025

Platform

- Designed a compact multi-sensor PCB in Altium Designer integrating: heart rate monitor, Time-of-Flight distance sensor (secondary daughter-board soldered perpendicularly), OLED display, STM32 microcontroller, USB-C power & communication, and battery charging.
- Implemented a 4th-order Butterworth low-pass filter using only COG/NPO components for clean analogue accelerometer signal conditioning — directly applicable to non-invasive sensing pipelines.
- Designed for miniaturisation: 0402 SMD components, stencil-applied solder paste, hand-soldered; replaced recommended Arduino shield with integrated STM32 for enhanced performance.

6-DOF Robotic Arm Simulation & Control

Distinction — Industrial Robotics 2025

Software

- Developed Python software to simulate and control three 6-DOF robotic arms, implementing forward/inverse kinematics and trajectory planning algorithms relevant to automated inspection pipelines.

Cart-Pendulum Balancing — Embedded Control

HD — Intro to Mechatronics 2024

System

Organised and delivered STEM Outreach program: physical double pendulum on cart system; UTS Robotics 2025

Programme chairing, governance, fundraising, coaching, operations, membership, and community services

LEADERSHIP & OUTREACH